5ELEN018W - Robotic Principles
Lecture 5: Control - Part 3

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Analysis of Control Systems - Performance of Controllers

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- steady-state error

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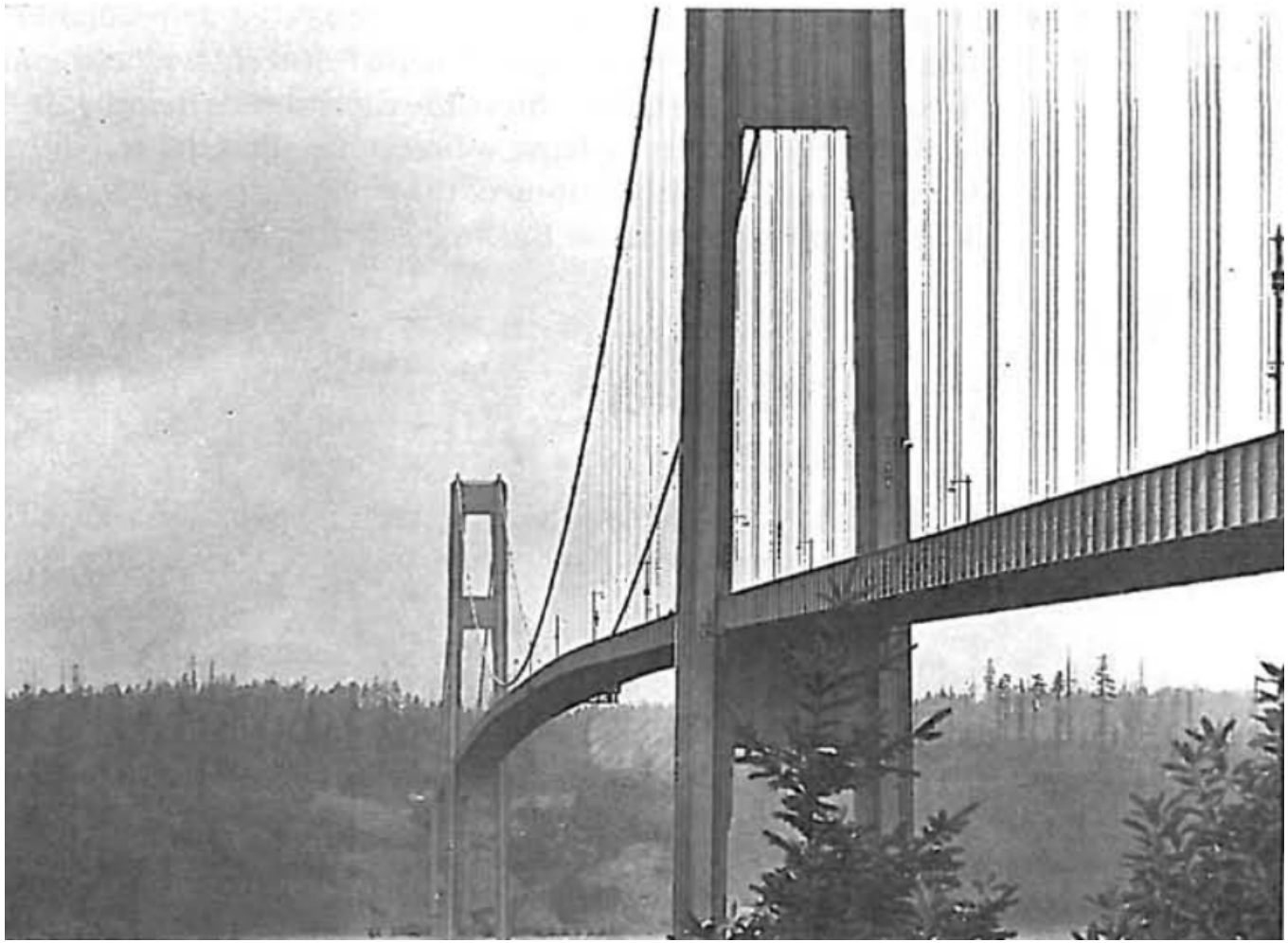
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https://www.youtube.com/watch?v=y0xohjV7Avo

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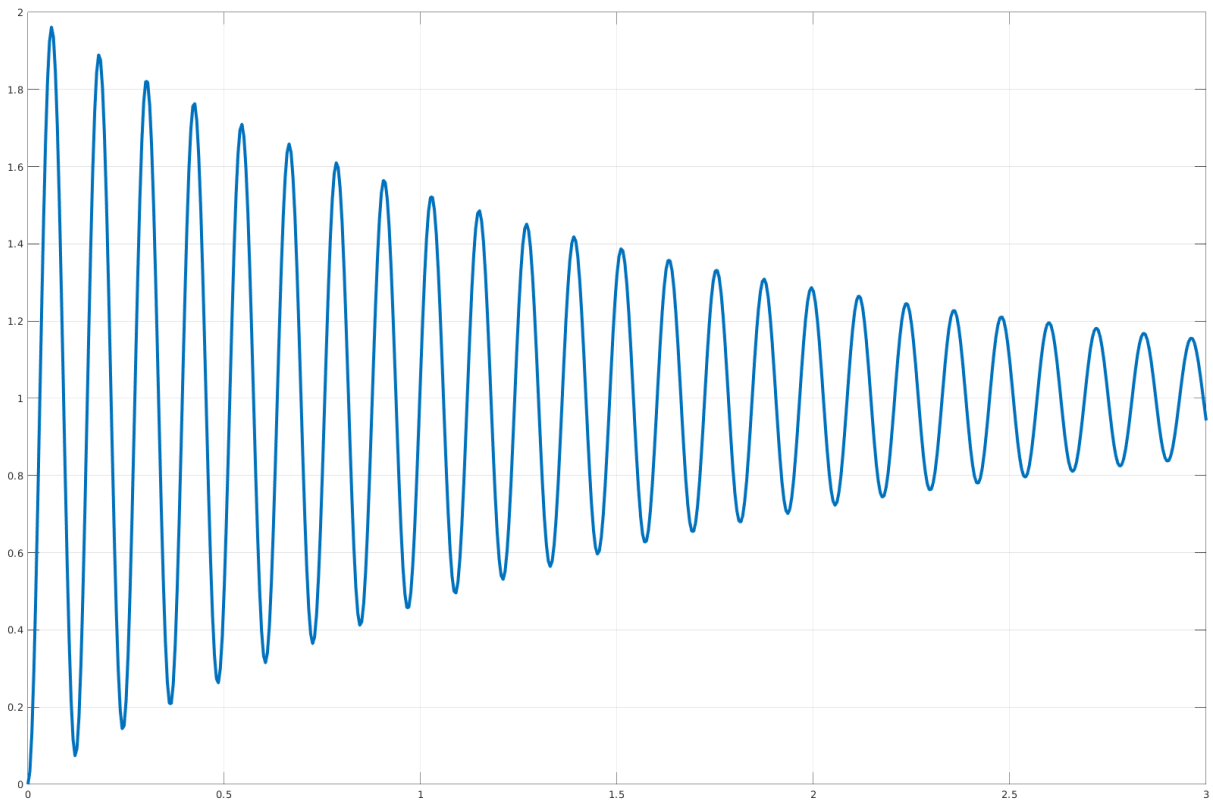
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Example of a Unit Step Response

System having the following transfer function:

$$\frac{5400}{2s^2 + 2.5s + 5402} \tag{1}$$



Example of a Unit Step Response (cont'd)

Matlab code:

```
num = [5400];
den = [2 2.5 5402];
sys = tf(num, den)
t = [0:0.005:3];
[y,t] = step(sys,t);
plot(t,y,'LineWidth',3),grid
xlabel('time(s)')
ylabel('System output')
```

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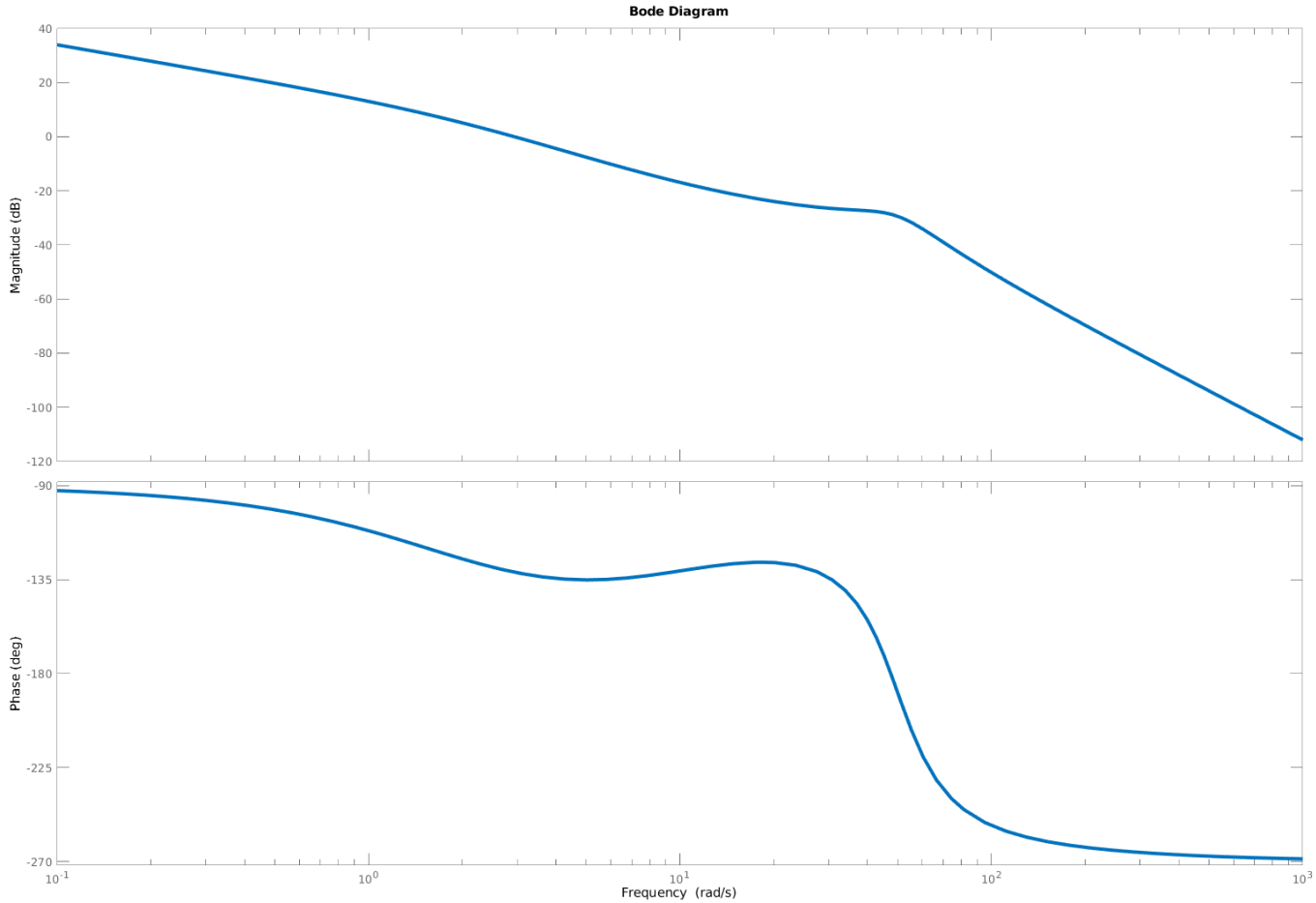
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- The phase plot (phase in degrees), frequency in rad/s.

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- The phase plot (phase in degrees), frequency in rad/s. Frequency values are in the logarithmic scale but the phase values are in the linear scale.

Example of a Bode Plot




```
sys=tf(num, den);   % previous system example
bode(sys)
```

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3. The **gain margin** is the difference between the magnitude of $0dB$ and M_1: $0 - M_1$.

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1. Find the frequency corresponding to a magnitude of $0dB$ from the magnitude plot.
2. For that frequency find the corresponding phase P_1 from the phase plot.
3. The **phase margin** is the difference between P_1 and -180°: $P_1 - (-180) = P_1 + 180$.

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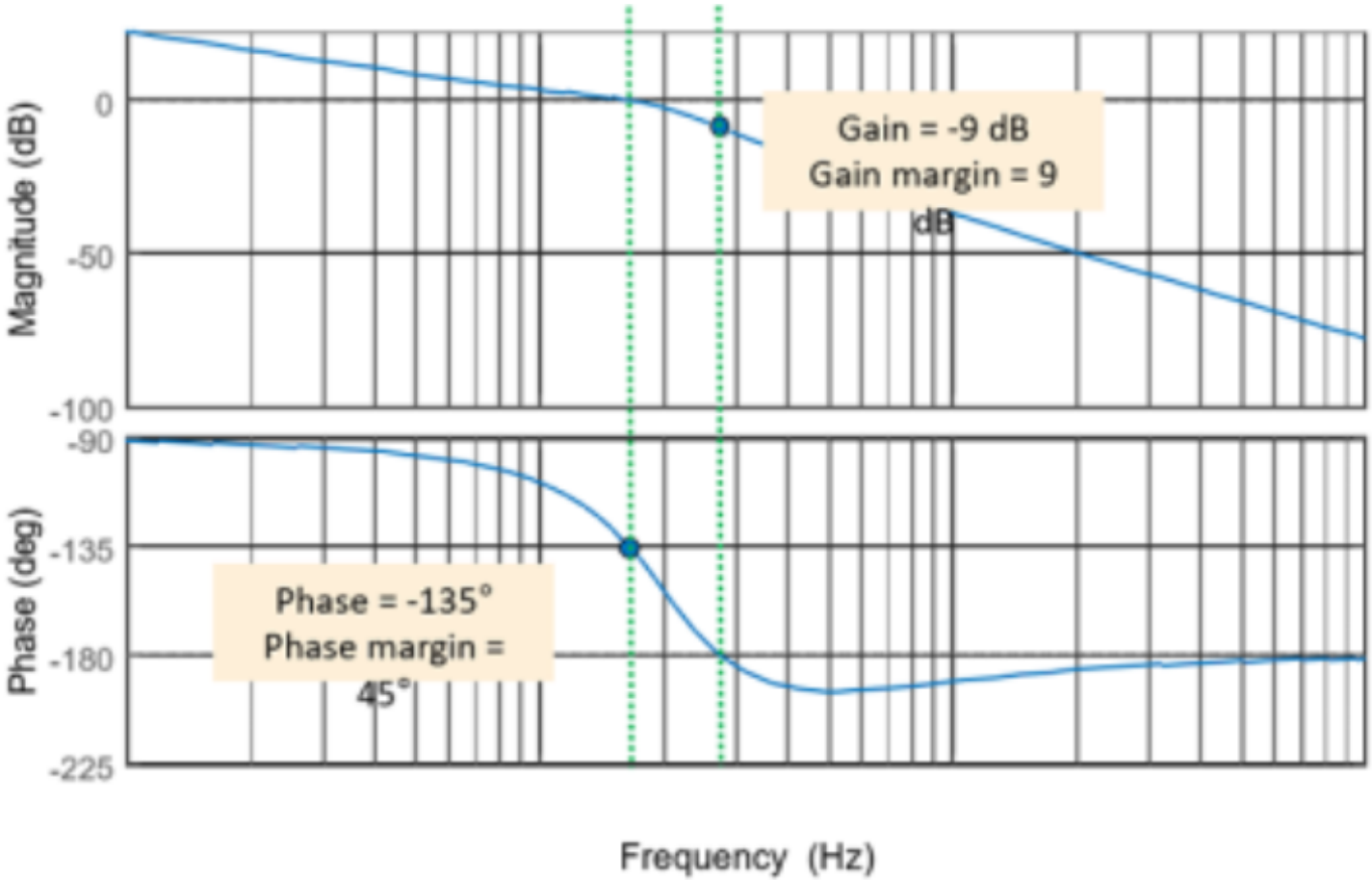
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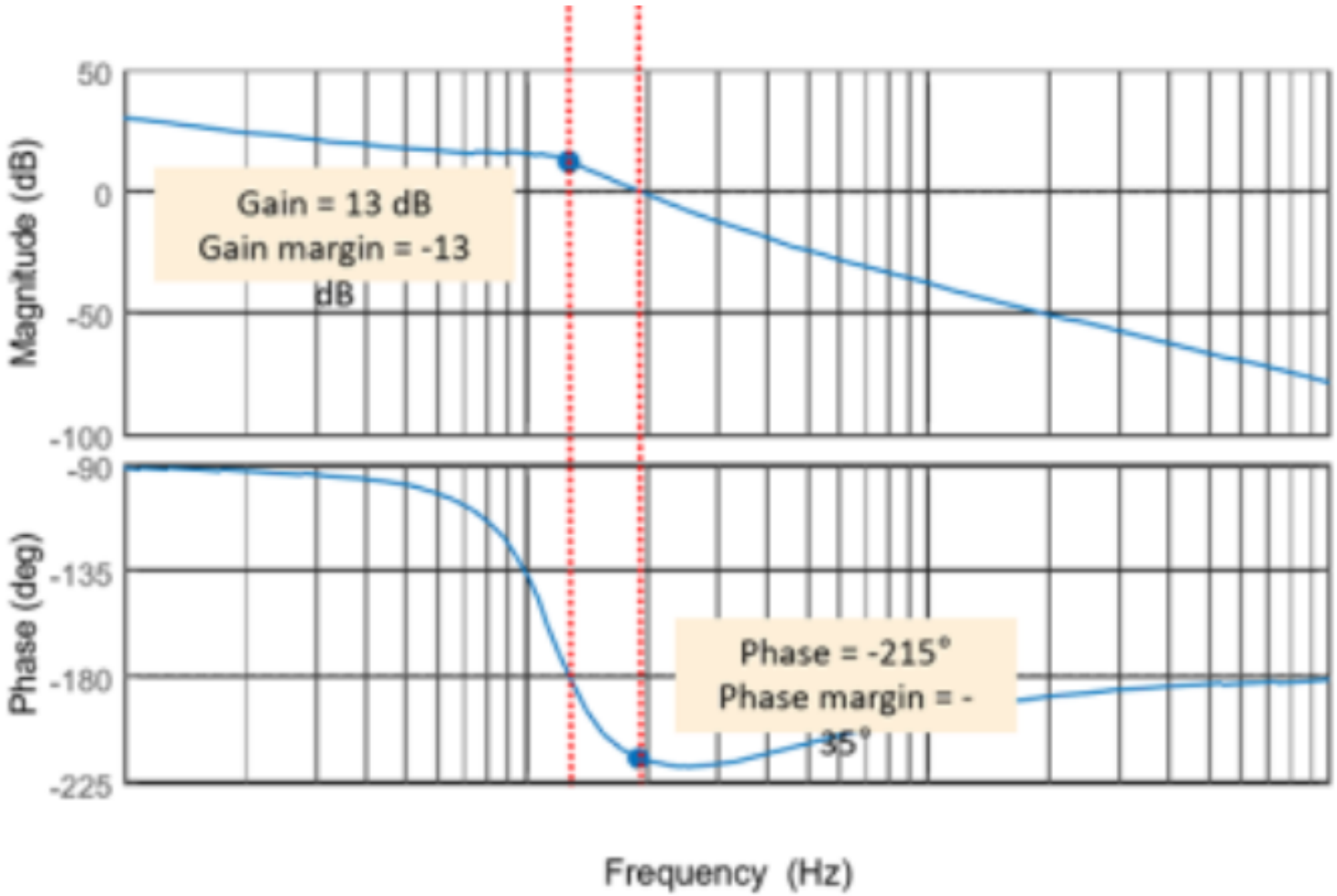
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- A positive gain margin indicates a stable system, while a negative gain margin indicates an unstable system. The larger the positive gain margin, the system has more "margin" before going to instability.
- A positive phase margin indicates a stable system, while a negative phase margin indicates an unstable system. The larger the positive margin the more stable is the system and a value of at least 45 degrees is desirable.

Example 2 of a Bode Plot



Example 3 of a Bode Plot



Bode Plots in Simulink

Problem: Watertank

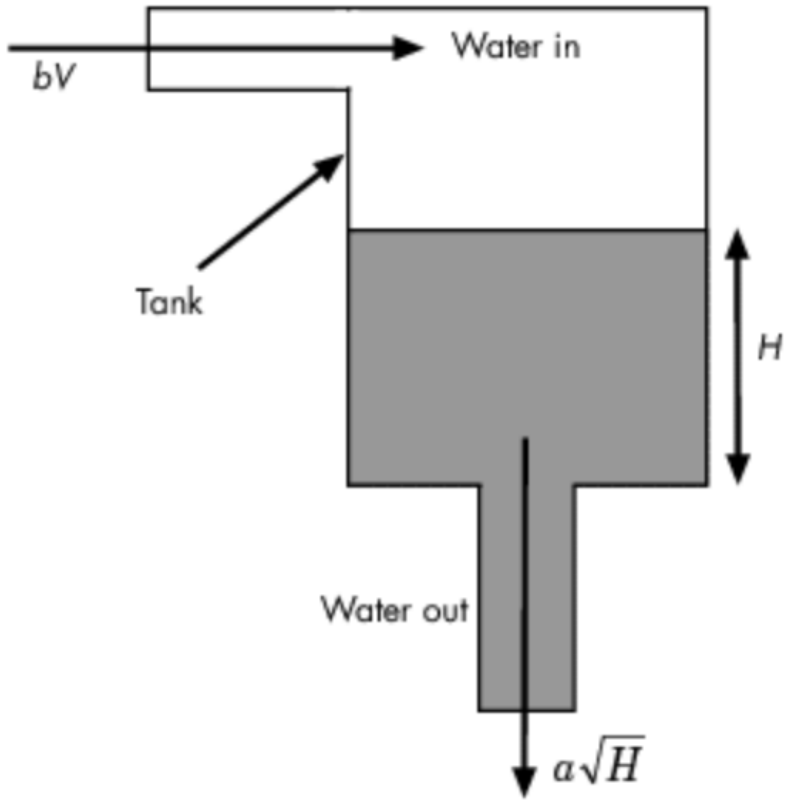


Figure 1: The watertank dynamic system.

Bode Plots in Simulink (cont'd)

$$\frac{d}{dt}Vol = A\frac{dH}{dt} = bV - \alpha\sqrt{H}$$

- H is the height of the water in the tank
- $Vol(V)$ is the volume of the water in the tank (control input)
- $b = 5$ is a constant related to the flow rate into the tank
- $A = 20$ relates to the area of the tank
- $\alpha = 2$ is a constant related to the flow rate out of the tank
- Initial condition for H is $H(0) = 1$

Bode Plots in Simulink (cont'd)

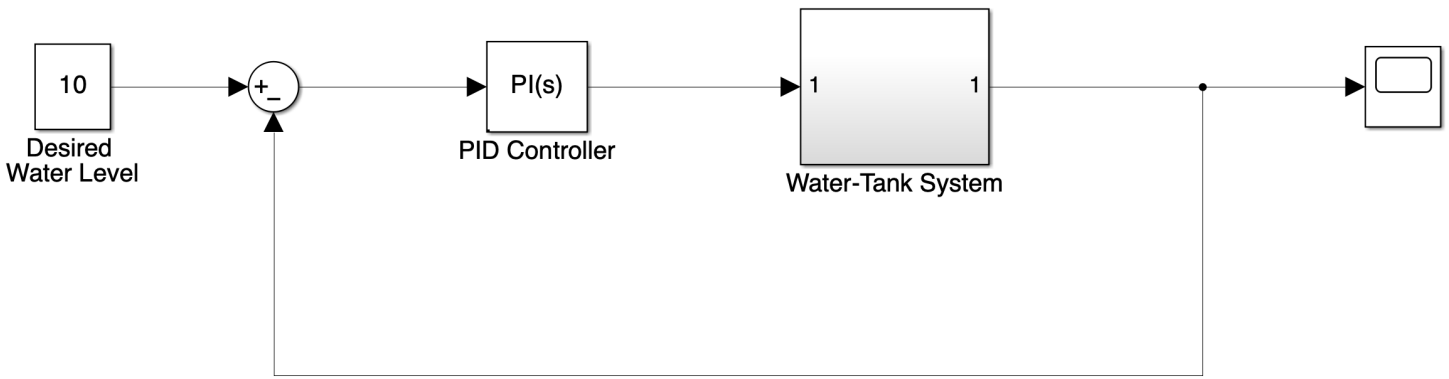


Figure 2: The watertank dynamic system in Matlab (built-in example up to version 2022).

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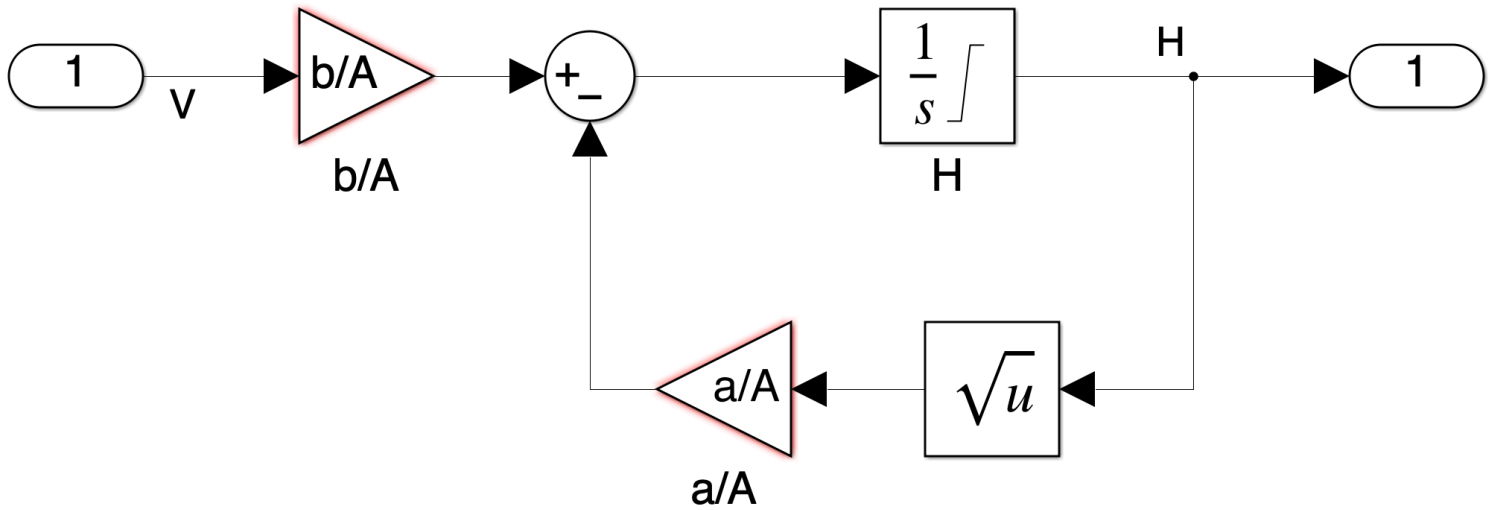


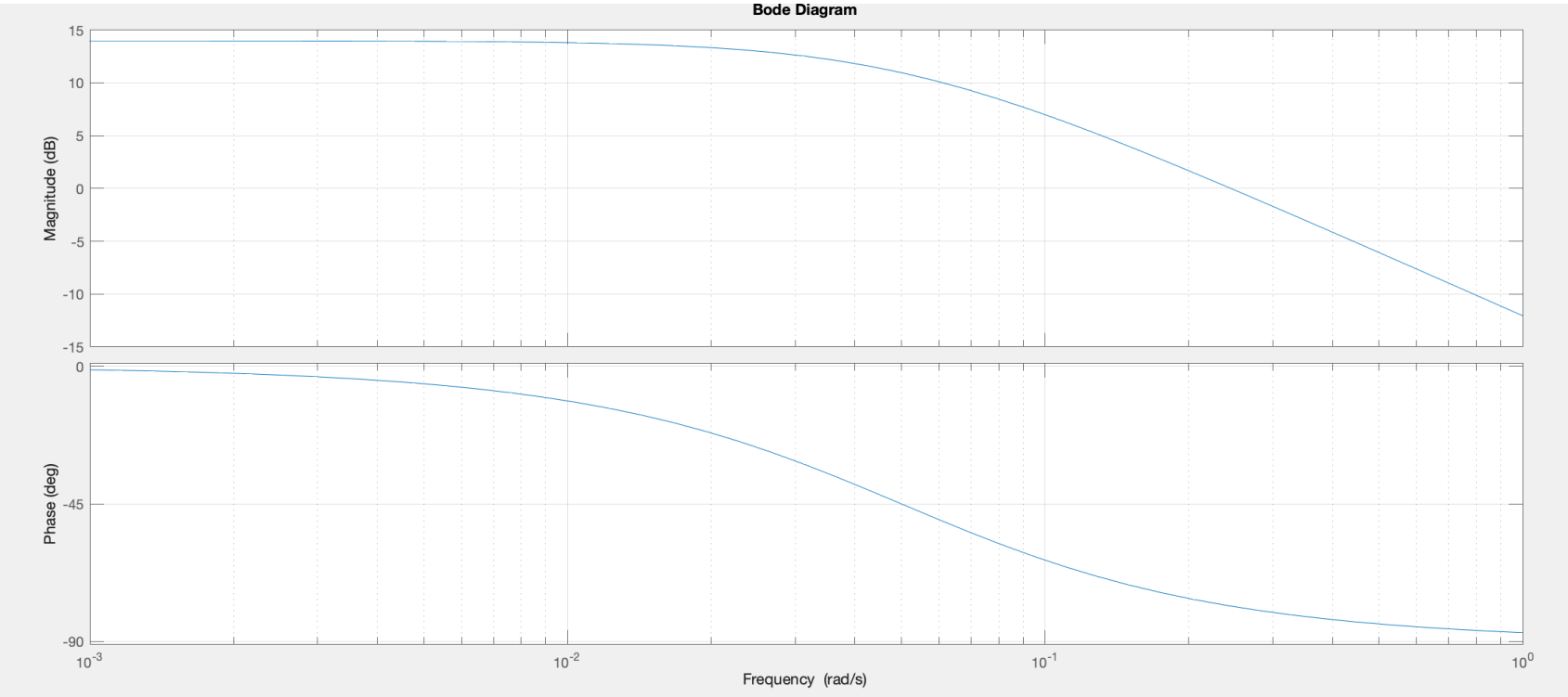
Figure 3: The watertank dynamic system in Matlab (built-in example up to version 2022).

Bode Plots in Simulink (cont'd)

From the Simulink Library Browser:

1. Choose Simulink Control Design
2. Choose Linear Analysis Plots
3. Choose and add a Bode Plot block
4. Open the Bode Plot block and in Linearisation add 2 signals:
   - → The output of the PID controller as input-perturbation
   - → The output of the plant as open-loop output
5. Run to display the Bode plot.

Bode Plots in Simulink (cont'd)



Implementing a Dynamic System and a Controller Programmatically (not Simulink)

Car cruise control can be found in most of the modern cars. The cruise control system keeps the car at a constant speed despite any external disturbances, such as the road surface and incline.

- The car's mass m is controlled by a force u. To simplify the situation, we assume that the force u that our controller applies is not affected by other parameters such as tires, etc.

The equation of the system is described by the following:

$$m\dot{v} + bv = u \tag{2}$$

where v is the speed of the car, u is the control action and b is the damping coefficient due to friction.

The Car Cruise Control System

$$m\dot{v} + bv = u \tag{3}$$

- $m = 1000, b = 50$.
- The desired (reference) speed is $v_{ref} = 10$.
- $t = 10.0$ is the total simulation time
- The parameters of the PID controller are:
  $K_p = 800, K_i = 0, K_d = 40$ (i.e. this is a PD controller).

Implement in Python a PID controller to bring the system to the desired speed.

The Car Cruise Control System (cont'd)

```
# Implements the dynamic system (plant) - system inputs are
# action_u and the current speed v.
# The function returns the output of the plant.
def plant(action_u, v):
    # equations of motion:
    # m \dot{v} + b v  = u
    # m=1000, b=50, u = 500
    m = 1000.0
    b = 50

    v_dot = (action_u - b*v)/m
    new_speed = v + v_dot*dt

    return new_speed
```

The Car Cruise Control System (cont'd)

```
v = 0  # current speed initialised to 0
previous_v = 0  # the speed at the previous time step
dt = 0.01  # time step for the simulation

start_time = 0.0
end_time = 10.0

current_time = start_time

v_ref = 10  # the desired speed

K_p = 800  # proportional gain
K_i = 41    # integral gain
K_d = 40   # derivative gain

previous_error = 0.0
integral = 0

# log values to file
f = open('myfile.txt', "w")
# write initial values to the file
f.write(f'{v}  {current_time}\n')
```

The Car Cruise Control System (cont'd)

```
    ''' Simulate the system operation from the beginning till
        the end of the simulation '''
    while current_time <= end_time:
        error = v_ref - v

        # I(integral) component of the PID controller
        integral = integral + error*dt

        # (D)erivative component of the PID controller
        deriv = (error - previous_error)/dt

        # the output (action) of the PID controller
        action = K_p*error + K_i*integral + K_d*deriv

        # remember the last error when the previous
        # action was applied to the plant
        previous_error = error

        # apply the new action to the plant to calculate the new (current) speed
        v = plant(action, v)

        print(f'Time: {current_time} Action: {action}, Speed={v}')
        f.write(f'{v}  {current_time}\n')

        # advance the time
        current_time += dt

        f.close()
```

Plotting the Response Program

File `cruisecontrol_plot.py`

```
import matplotlib.pyplot as plt

f = open('myfile.txt')
v = []
time =[]
for i in f:
    [y, t] = i.split()
    #print(y, t)
    v.append(float(y))
    time.append(float(t))

print(v)
plt.plot(time, v)
plt.axis([-1, 10, 0, 12])
plt.xlabel('Time t')
plt.ylabel('Speed v')
plt.show()
```

Running the Plot Program

